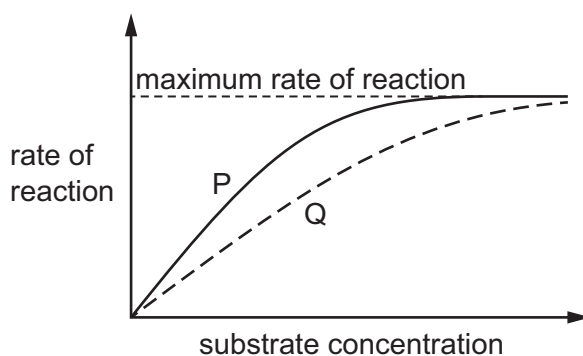


- 13 The graph shows the effect of an increasing substrate concentration on the rate of an enzyme-catalysed reaction.



Line P represents the result when the enzyme is used at its optimum pH and optimum temperature and without an inhibitor.

Line Q represents the result when the reaction conditions are changed.

Which descriptions of changes to the reaction conditions could result in line Q if all other conditions were kept the same?

- 1 Add an inhibitor that attaches to a site other than the active site.
- 2 Add an inhibitor that has a similar shape to the substrate.
- 3 Add an inhibitor that blocks the active site of the enzyme.
- 4 Carry out the reaction at a higher temperature.

A 1, 3 and 4 **B** 1 and 4 only **C** 2, 3 and 4 **D** 2 and 3 only